Po Jian

Senior Data Engineer: building data environments people can trust and actually use

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SUMMARY

- Data engineer with 5 years of experience in data analysis and data engineering; master's in Industrial and Organizational Psychology.
- Led data warehouse design, an in-house event-tracking tool, an Airflow migration to Kubernetes, a data analysis white paper, and an internal data portal at BenQ.
- Values evidence-driven analysis, empathy when a team changes how it works, and putting data in users' hands.

SKILLS

- **Languages:** Python, SQL
- **Data engineering:** Airflow, PySpark, Kafka, Hadoop, Hive
- **Warehouse / platform:** Databricks, BigQuery, OpenSearch/ELK, MySQL
- **Data governance:** DataHub, data modeling, documentation, ownership practices
- **Cloud:** AWS (S3, SQS, Athena, QuickSight), GCP
- **Infrastructure / monitoring:** Kubernetes, Docker, Linux, Prometheus, Grafana
- **Other:** API platform development, Tableau

EXPERIENCE

Senior Data Engineer | BenQ — May 2024 – Present

- Built **Momotaro**, an in-house event-tracking tool that replaced a legacy Matomo setup on EC2 without requiring app changes (Python, Flask, microservices, AWS).
- Migrated Airflow from the Celery executor to the Kubernetes executor; set up Prometheus + Grafana monitoring and rebuilt backups.
- Designed the data warehouse layers and access-control model; set team documentation standards.
- Started a project Owner system and a weekly study group.
- Built a data analysis white paper using Databricks Genie (Text-to-SQL) and an internal Insight Portal.
- Developed a shared API platform and a Data QA automation tool (Appium) that reduces analysts' manual work.
- Day to day: refactored existing ETL, operated the related infrastructure, onboarded third-party data sources.

Data Engineer | Pixl Solutions — Aug 2023 – Apr 2024

- Managed and optimized data warehouses on AWS and GCP.
- Operated and troubleshot ELK services (Elasticsearch, Logstash, Kibana).
- Supported cross-department data requests and analysis.

Part-time Teaching Assistant | Alpha Camp (career break) — Apr 2023 – Jul 2023

- Supported live sessions, reviewed assignments, and answered questions for SQL and data courses.

Data Engineer | KKStream — Aug 2022 – Apr 2023

- Built an internal data-processing framework and AWS integrations.
- Refactored a performance-reporting module and designed its metrics.
- Designed dashboards and data visualizations.

Data Engineer | Gamania — Jun 2021 – Aug 2022

- Maintained, optimized and built ETL pipelines.
- Designed a marketing-ad data warehouse and its API integrations.
- Led a data dictionary project with DataHub.
- Built monitoring dashboards in Grafana.

Smart Education Administrator | National Yunlin University of Science and Technology — Mar 2019 – May 2021

- Developed and trained a neural network model for learning-behavior recommendations.
- Built crawlers, notification programs and a LINE bot; designed and analyzed notification-style experiments.
- Integrated and maintained a face-recognition system.
- Analyzed learning behavior on digital platforms; designed eye-tracking and cognitive-psychology experiments.
- Designed a smart classroom (behavior labels, indicators, reports) and improved office processes.

Data Analyst Intern / Assistant | KKday; Hongguang Data — Jul 2017 – Jun 2018

- KKday: maintained a daily price-comparison reporting system with automated report emails, and its crawler.
- Hongguang Data: wrote web crawlers in R and helped clean data; assisted with semantic-system training.

SELECTED PROJECTS

Stark Support Bot | LLM guardrails and evaluation (personal project) Oct 2026

- RAG customer-service bot used as a test bed: started defenseless, added one defense at a time, and re-ran the same 28 attacks and 6 normal questions after each step.
- Attacks blocked rose from 16/28 to 28/28 while normal questions stayed 6/6 answerable.
- Method: deterministic canary and doc-fact checks (no LLM judge), confidence intervals on small samples, 143 labelled classifier cases (137 correct), documented limits.
- Model failures never count as blocked attacks; the bot fails closed and escalates to a human.
- Task requests dressed up as product questions are treated as off-topic (found with adversarial test questions).
- Compared embedding models on 90 retrieval questions (hit@3, MRR) and chose bge-m3 for Chinese queries.
- Server keeps the conversation state, so a browser cannot forge earlier turns; tickets and MLflow traces carry the session id.
- Python, FastAPI, MLflow, bge-m3 retrieval.

agent-perch | open-source supervisor for long-running AI agents (MIT) Oct 2026

- Single-file, standard-library-only Python tool that keeps agent processes (e.g. Claude Code bots) alive.
- Restarts dead sessions, answers known dialogs, detects frozen prompts, and halts crash loops with alerts.
- File-based mailbox for agent-to-agent messages and a small dashboard.
- Crash-loop halt (3 restarts in 10 minutes); alerts go through a user-supplied command (Telegram, Slack, mail).
- Dashboard controls are token-protected.
- Generalized from a watchdog that ran a live Raspberry Pi agent fleet; every design decision in the README states its trade-off.

Data Ingestion Platform | personal project Mar 2025

- Web interface for departments without technical staff to contribute data as CSV uploads.
- Validates each upload against a per-table schema maintained by the data team and shows a preview.
- Reports success or failure clearly; Docker Compose setup. Limits documented (proof of concept).

YouBike Historical Data | personal project 2025

- Collects Taipei YouBike availability every ten minutes into a fact table and a station dimension table.
- Publishes daily data files with a documented schema and known data pitfalls; unit tests and alerting.
- Chose the station's upload time over the system receive time as the timestamp, with the reasoning documented.

WRITING

- **The Evolution of Our Data Team** (Medium, Chinese and English): a series on how a data team grows.
- Culture: the Owner system.
- Technical: serving API platform, event-tracking tool and QA framework, migrating Airflow to Kubernetes, system operations.
- AI: data insight reports (strategy and implementation) and the Insight Portal.
- Also work notes and reading notes.

EDUCATION

- **M.A., Industrial and Organizational Psychology**, National Chengchi University (Sep 2016 – Mar 2019)
- **B.A., Psychology**, National Chung Cheng University (Sep 2012 – Jun 2016)

CERTIFICATIONS AND TRAINING

- Databricks Certified Data Engineer Associate (May 2026)
- AWS Certified Data Engineer – Associate (Feb 2024)
- Google Cloud Associate Cloud Engineer (Apr 2023)
- Coursera AWS Fundamentals specialization, 4 courses (Sep 2022)
- iPAS Junior Big Data Analyst (Dec 2018); BI and Big Data Analyst training, 261 hours (2018)